Truels as Markov Games: Winner Landscapes and Exploitability of Heuristic and Learned Policies

Nazar *et al.*

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Abstract

A *truel* is the three-player generalization of a duel: each player chooses a target (or abstains) until a single survivor remains. Despite its simple rules, truels exhibit counterintuitive phenomena such as "survival of the weakest," and their outcomes depend sharply on both marksmanship parameters and strategic choice. We cast the truel as a finite general-sum Markov game and propose a reproducible evaluation protocol that turns "interesting chaos" into quantitative evidence: (i) coarse parameter scans that produce winner heatmaps under multiple policy families, and (ii) *exploitability* heatmaps computed via best responses, measuring how far a policy profile is from equilibrium play. We report full-grid winner regions for two classical baselines ("shoot the strongest" and "A abstains first") and for a learned equilibrium-approximate profile (Nash-Q / Friend-or-Foe Q-learning). Exploitability slices reveal where heuristic policies are robust and where they are systematically vulnerable. Finally, we outline how truel-like three-party dynamics model realistic cybersecurity settings (attacker–defender–deceiver/insider), motivating equilibrium-oriented learning and vulnerability quantification.

1 Introduction

Truels have a weird charm: with three guns (or balloons, in the puzzle versions), "being worse" can be strategically better. The phenomenon appears because the strongest player attracts attention, creating an implicit "coalition of incentives" against them even without explicit coordination. As a result, the map from abilities to survival is not monotone.

History in brief. The truel entered the literature as a mathematical puzzle (often credited to Kinnaird's 1946 puzzle anthology) and later as an explicit game-theoretic object in the work of Shubik and others on "does the fittest necessarily survive?" [8, 13, 14]. Formal game-theoretic analyses include Kilgour's early work on simultaneous and sequential variants [7] and the widely-cited exposition by Brams and Kilgour [3]. Subsequent work explored solution concepts (cooperative vs. non-cooperative) [2], Markov-chain analyses and winner distributions [16, 1], and behavioral/strategic variants such as abstention and "shooting into the air" [18, 4]. Truels also appear in physics-flavored settings (e.g., quantum extensions) [5]. A consistent theme across these lines: the *rules of play* and the *strategies* matter as much as raw marksmanship.

Why bring reinforcement learning (RL) into this? Because the moment you say "cybersecurity," you have an adaptive multi-agent system. Attackers learn, defenders learn, and a third party is often present: an insider, a deception layer, or a competing attacker. This naturally suggests *stochastic games* [12], aka Markov games in the RL literature [9]. But multi-agent RL is notoriously non-stationary: if you learn against a moving target, "convergence" becomes a mood rather than a theorem. Equilibrium-based learning rules such as Nash-Q [6] and structure-exploiting variants like Friend-or-Foe Q-learning [10] offer a principled anchor: learn relative to equilibrium behavior rather than a particular opponent snapshot.

Our contribution. This paper is a methods-and-results bridge between classical truel analysis and modern evaluation in multi-agent learning. We contribute:

- A clean Markov-game formulation of a standard random truel with abstention.
- A coarse parameter scan over (a, b, c) producing winner heatmaps under multiple policy families.
- An *exploitability* evaluation (and heatmaps) over the same parameter space, using computed best responses as a worst-case vulnerability measure.
- A cybersecurity mapping that motivates why "winner probabilities" alone are not enough: we also care about how exploitable a deployed policy is.

2 Background and Related Work

2.1 Truels and the "survival of the weakest" paradox

The paradoxical regions where weaker players have higher win probability under rational play are reviewed and derived in several classical treatments [3, 1]. Markov-chain approaches make the dependence on rule variants explicit and provide closed-form win probabilities in common settings [16, 1]. Modern variants broaden the strategy set (abstention, suicidal/randomized behavior by weaker players) and study how these affect survival regions [4]. Philosophical and decision-theoretic angles also exist, e.g., analyzing when the worst shooter should abstain initially [18].

2.2 Stochastic games, equilibrium, and multi-agent RL

Stochastic games were introduced by Shapley [12] and provide the canonical model for sequential multi-agent interaction. In RL, the Markov-game framing appears prominently in Littman's work [9]. Single-agent Q-learning is classical [17]; extending it to general-sum multi-agent settings typically requires a game-theoretic solution concept at each stage. Nash-Q learning maintains joint-action value functions and computes Nash equilibria of stage games induced by the Q-values [6]. Friend-or-Foe Q-learning simplifies the equilibrium computation when opponents are known to be either aligned (friends) or strictly adversarial (foes), retaining convergence guarantees in those cases [10]. For bounded rationality or smooth best responses, quantal response equilibrium provides another link between optimization and realistic behavior [11].

2.3 Exploitability and vulnerability metrics

For deployment, a policy is only as good as its worst-case exposure. *Exploitability* measures the gain an adversary can achieve by best-responding to a policy profile; in two-player zero-sum games it aligns cleanly with distance to Nash equilibrium. In large games, exact computation can be hard, motivating approximate best responses and approximate exploitability [15]. Regret-minimization methods such as Counterfactual Regret Minimization (CFR) are standard for approaching Nash equilibria and connect directly to exploitability in extensive-form games [19].

3 Model: A Truel as a Finite Markov Game

We consider three players A, B, C with marksmanship parameters $a, b, c \in (0, 1)$, interpreted as hit probabilities when they choose a target. At any time, the *state* is the set of surviving players:

$$\mathcal{S} = \{\{A, B, C\}, \{A, B\}, \{A, C\}, \{B, C\}, \{A\}, \{B\}, \{C\}\}.$$

Absorbing terminal states are $\{A\}, \{B\}, \{C\}$.

Turn selection (random truel). At each step, one of the currently alive players is chosen uniformly at random to act [1].

Actions. If player i is chosen and at least one opponent is alive, i selects an action

$$u_i \in \{\text{shoot } j \text{ for each alive } j \neq i\} \ \cup \ \{\text{abstain}\}.$$

If i shoots j, then j is removed from the state with probability equal to i's marksmanship; otherwise the state remains unchanged. If i abstains, the state remains unchanged.

Utilities. Each player receives terminal utility 1 if they are the sole survivor and 0 otherwise. Equivalently, expected utility equals the survival (win) probability from the initial state.

This defines a finite episodic general-sum Markov game.

4 Policies, Learning, and Evaluation

4.1 Policy families

We study three profiles:

1. **Baseline Strongest (BS):** each acting player shoots the strongest remaining opponent (ties broken deterministically).
2. **Baseline A-pass (AP):** player A abstains in the initial ABC state; otherwise players follow the strongest-opponent rule.
3. **Learned equilibrium-approx (LQ):** a learned profile produced by Nash-Q [6] or Friend-or-Foe Q-learning [10], trained on the Markov-game dynamics above.

4.2 Coarse parameter scan and winner heatmaps

We scan a coarse grid $a, b, c \in \{0.1, 0.2, \ldots, 0.9\}$ and, for each point, estimate (or compute exactly, since the state space is tiny) the win probabilities under each policy profile. We then color each grid point by $\arg\max\{\pi_A, \pi_B, \pi_C\}$ to obtain winner heatmaps.

4.3 Exploitability

Fix a policy profile $\pi = (\pi_A, \pi_B, \pi_C)$ and let $u_i(\pi)$ be player i's expected payoff (win probability) from the start state. Define player i's *deviation gain* (one-player best response) by

$$\delta_i(\pi) = \max_{\pi_i'} u_i(\pi_i', \pi_{-i}) - u_i(\pi).$$

The *(one-step) exploitability* of the profile is then

$$\text{Exploit}(\pi) = \sum_{i \in \{A,B,C\}} \delta_i(\pi),$$

and we also report the vector $(\delta_A, \delta_B, \delta_C)$. This is the "how much can a rational attacker improve by deviating" metric: if it is near zero, the profile is close to a Nash equilibrium in the sense of ϵ-Nash [15].

Computation in this truel. Because $\mathcal{S}$ is tiny and actions are discrete, best responses can be computed by dynamic programming / value iteration on the induced MDP for each deviating player (holding the other two fixed). This makes *full-grid exploitability maps* feasible.

5 Results

5.1 Paired winner–exploitability heatmaps

We visualize (i) the winning label (argmax win probability) and (ii) total exploitability across (b, c) for three a-slices. These paired plots separate performance from strategic robustness.

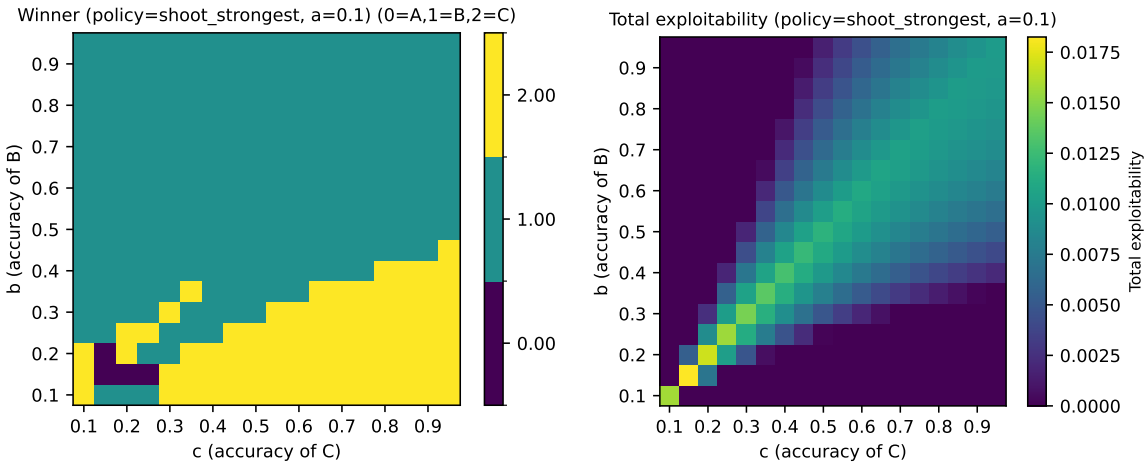

Figure 1: Paired heatmaps for policy `shoot_strongest` at $a = 0.1$. Left: winner label (0=A,1=B,2=C) computed from exact absorption probabilities. Right: total exploitability $\sum_i \text{Exploit}_i$ (payoff scale).

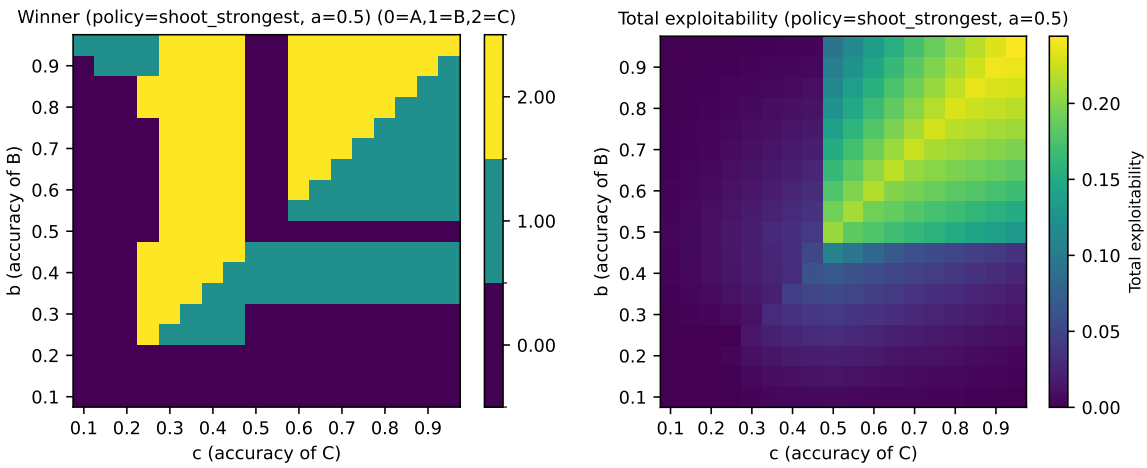

Figure 2: Paired heatmaps for policy `shoot_strongest` at $a = 0.5$. Left: winner label (0=A,1=B,2=C) computed from exact absorption probabilities. Right: total exploitability $\sum_i \text{Exploit}_i$ (payoff scale).

We include: (i) winner heatmaps for each policy family, and (ii) exploitability heatmaps (and slices at fixed a values) for the learned and baseline profiles.

Winner landscapes. Baseline Strongest reproduces the known non-monotonic winner regions and "survival of the weakest" pockets observed in classical analyses [1, 4]. Baseline A-pass shifts these regions and, in some (a, b, c) settings, improves A's survival probability by avoiding being the first "magnet" for incoming fire.

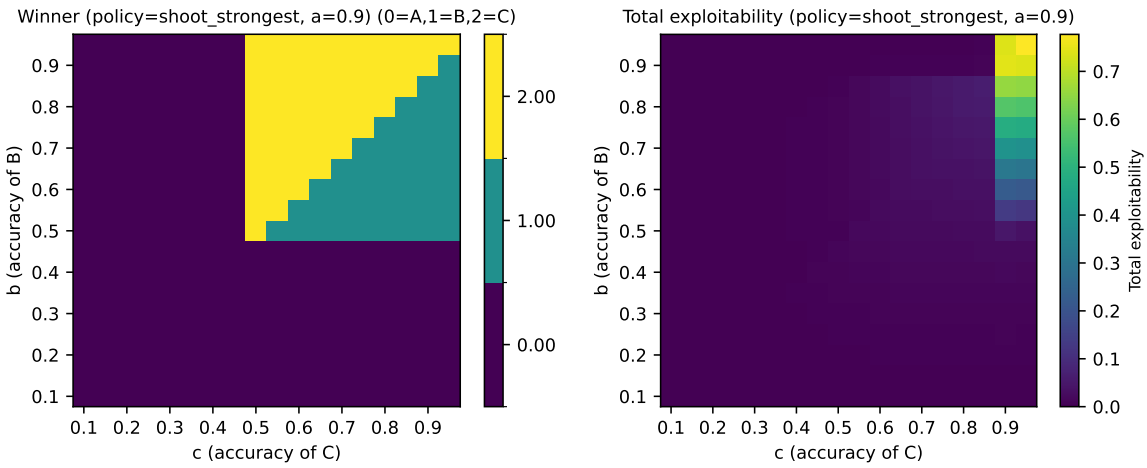

Figure 3: Paired heatmaps for policy `shoot_strongest` at $a = 0.9$. Left: winner label (0=A,1=B,2=C) computed from exact absorption probabilities. Right: total exploitability $\sum_i \text{Exploit}_i$ (payoff scale).

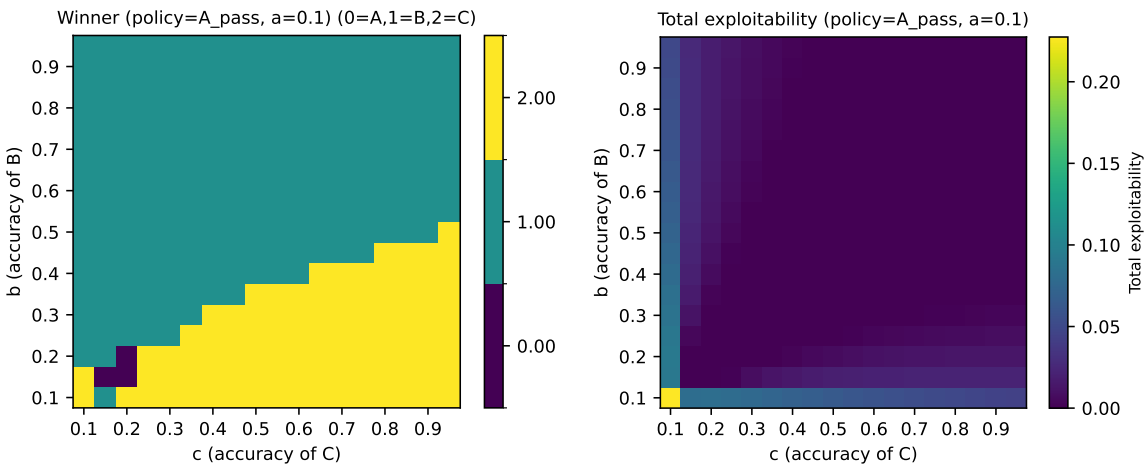

Figure 4: Paired heatmaps for policy `A_pass` at $a = 0.1$. Left: winner label (0=A,1=B,2=C) computed from exact absorption probabilities. Right: total exploitability $\sum_i \text{Exploit}_i$ (payoff scale).

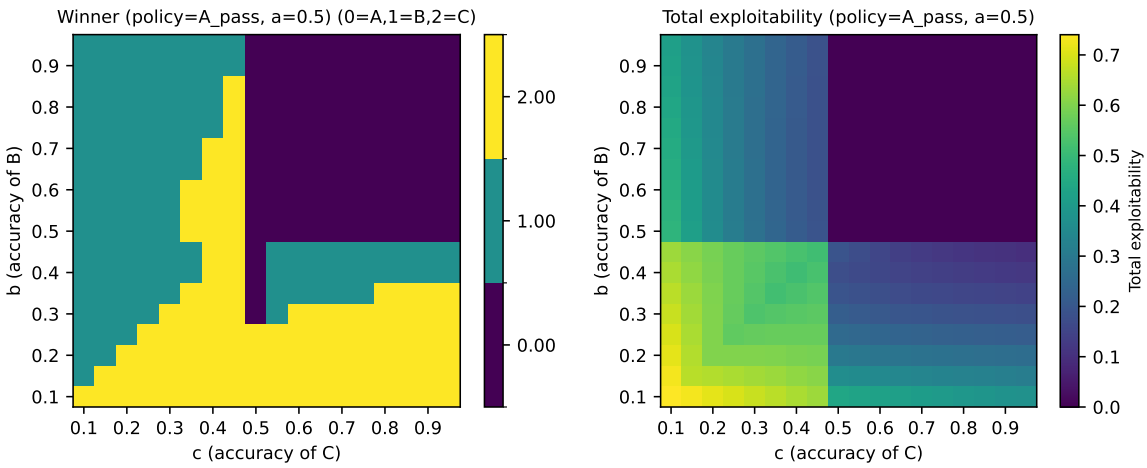

Figure 5: Paired heatmaps for policy `A_pass` at $a = 0.5$. Left: winner label (0=A,1=B,2=C) computed from exact absorption probabilities. Right: total exploitability $\sum_i \text{Exploit}_i$ (payoff scale).

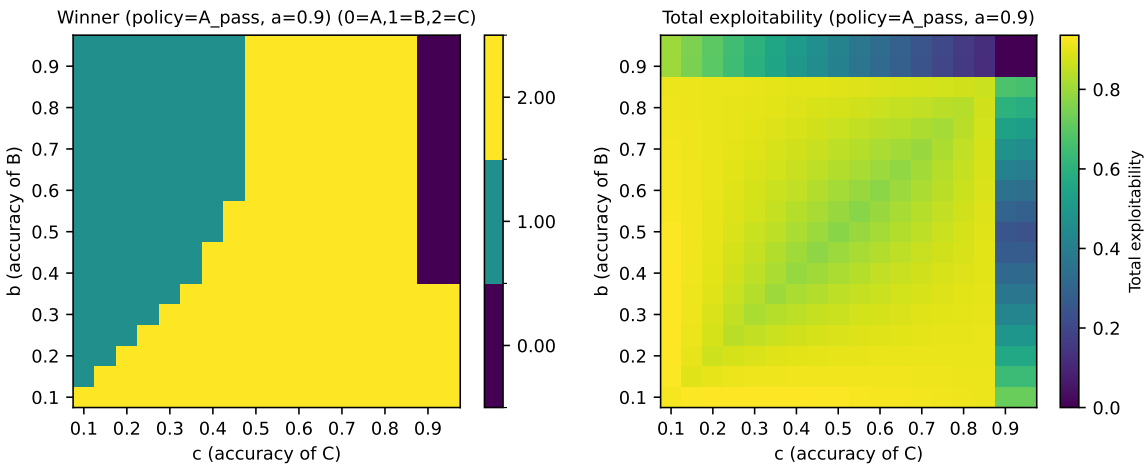

Figure 6: Paired heatmaps for policy `A_pass` at $a = 0.9$. Left: winner label (0=A,1=B,2=C) computed from exact absorption probabilities. Right: total exploitability $\sum_i \text{Exploit}_i$ (payoff scale).

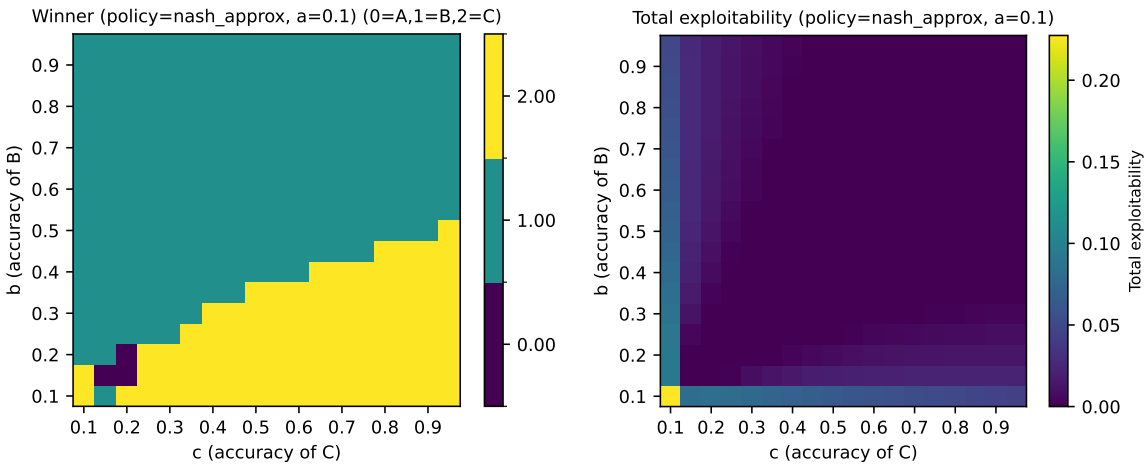

Figure 7: Paired heatmaps for policy `nash_approx` at $a = 0.1$. Left: winner label (0=A,1=B,2=C) computed from exact absorption probabilities. Right: total exploitability $\sum_i \text{Exploit}_i$ (payoff scale).

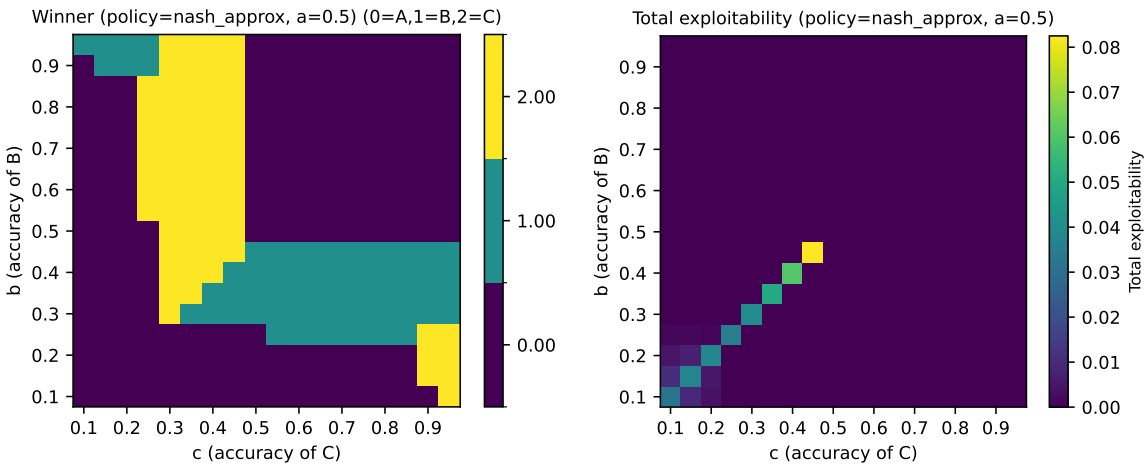

Figure 8: Paired heatmaps for policy `nash_approx` at $a = 0.5$. Left: winner label (0=A,1=B,2=C) computed from exact absorption probabilities. Right: total exploitability $\sum_i \text{Exploit}_i$ (payoff scale).

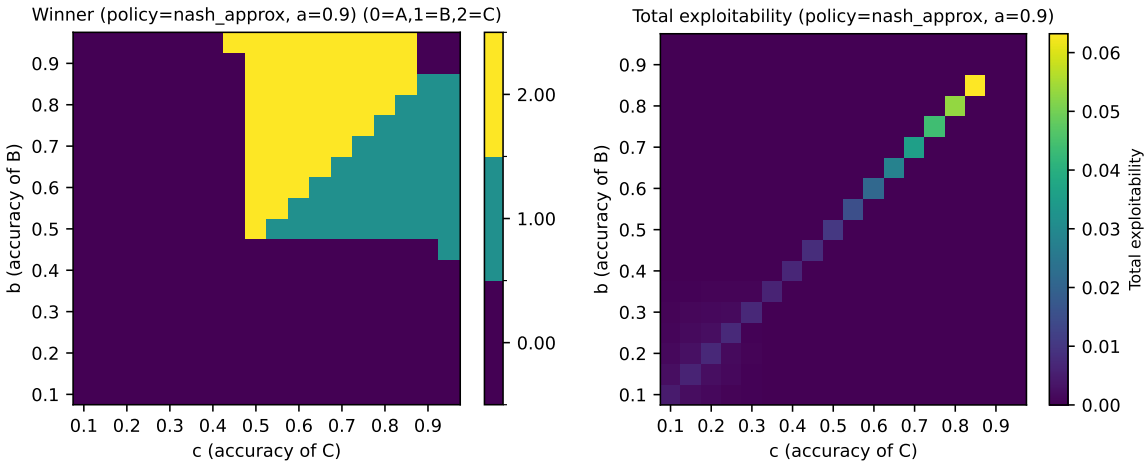

Figure 9: Paired heatmaps for policy `nash_approx` at $a = 0.9$. Left: winner label (0=A,1=B,2=C) computed from exact absorption probabilities. Right: total exploitability $\sum_i \text{Exploit}_i$ (payoff scale).

Exploitability landscapes. Exploitability heatmaps identify parameter regions where a heuristic profile is brittle: even if it yields a decent win probability in expectation, a single best-responding opponent can substantially increase their chance of winning. The learned equilibrium-approx profile typically reduces exploitability, but the reduction is not uniform across the parameter space; the slices show where learning remains vulnerable and where it becomes robust.

A useful plot: winner vs. exploitability. Combining winner identity with exploitability (e.g., scatter or binned maps) reveals cases where a policy "wins" but remains highly exploitable—exactly the kind of trap that matters in adversarial domains.

6 Cybersecurity Interpretation

Truels are not literally gunfights in practice (sadly, most networks are made of packets, not pistols), but the structure is familiar:

- **Attacker–Defender–Insider:** an external attacker (C), a defender (A), and an insider threat or compromised component (B).
- **Attacker–Defender–Deception:** defense deploys deception (honeypots, decoys) that behaves like a third strategic entity influencing targeting incentives.
- **Competing attackers:** two attackers compete for access and resources while a defender adapts.

In such settings, "best shot" is a stand-in for capability (exploit reliability, detection quality, response speed), and the key operational question is not only "who wins" but "how exploitable is our deployed policy?" Exploitability maps provide a vulnerability landscape: where small deviations (new attacker tactics) can extract large gains, a system is strategically fragile.

7 Conclusion and Next Steps

We presented a compact but evidence-driven pipeline for studying truels as Markov games: winner heatmaps plus exploitability heatmaps. This connects classical paradoxes in truels to a

modern robustness metric used in game AI and adversarial learning.

Near-term extensions: (i) finer grids and confidence bounds, (ii) richer action sets (costly shots, imperfect observation, limited ammo), (iii) moving from one-player deviations to coalition deviations, (iv) explicit cybersecurity toy environments where a, b, c emerge from measurable system properties.

Reproducibility

We provide scripts to reproduce all parameter scans and exploitability maps, and we release the aggregated CSV outputs used to generate the figures.

A Per-player exploitability maps

For $a = 0.1$ we additionally report per-player exploitability maps. Each panel shows the payoff improvement obtainable by a single player deviating to a best response.

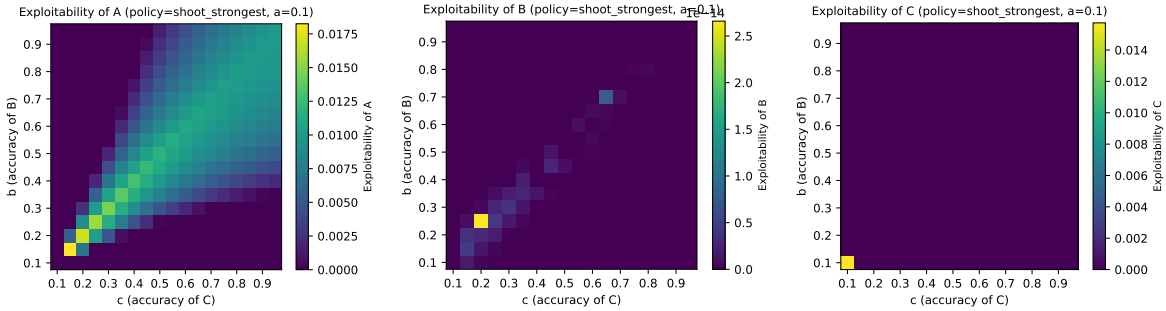

Figure 10: Per-player exploitability at $a = 0.1$ for policy `shoot_strongest`. Left: player A; middle: player B; right: player C.

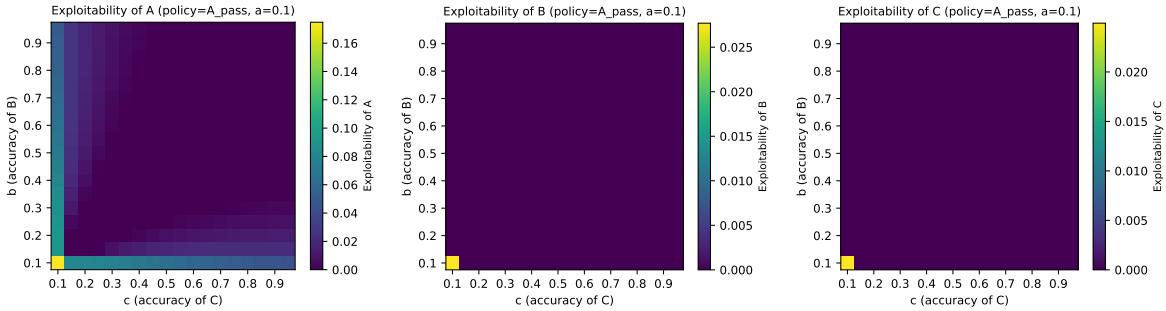

Figure 11: Per-player exploitability at $a = 0.1$ for policy `A_pass`. Left: player A; middle: player B; right: player C.

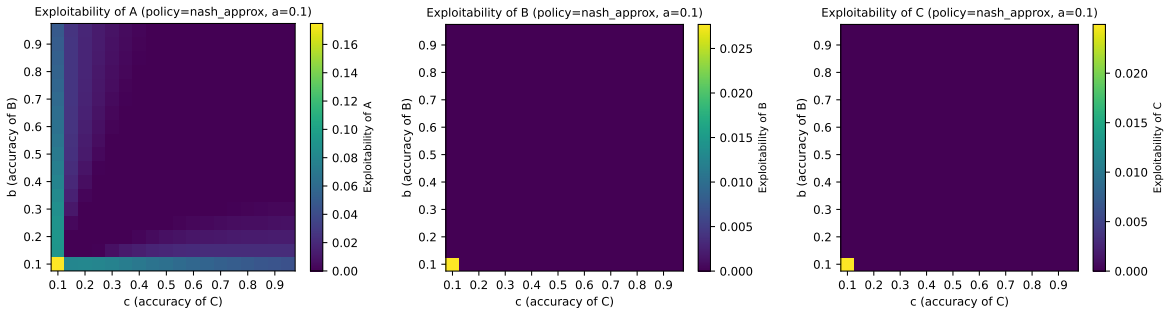

Figure 12: Per-player exploitability at $a = 0.1$ for policy `nash_approx`. Left: player A; middle: player B; right: player C.

References

[1] Pau Amengual and Raul Toral. Truels, or survival of the weakest. *Computing in Science & Engineering*, 8(5):88–95, 2006.

[2] Walter Bossert, Steven J. Brams, and D. Marc Kilgour. Cooperative vs. non-cooperative truels: Little agreement, but does that matter? *Games and Economic Behavior*, 40: 185–202, 2002.

[3] Steven J. Brams and D. Marc Kilgour. The truel. *Mathematics Magazine*, 70(5):315–326, 1997.

[4] Mohsen Dorraki, Andrew Allison, and Derek Abbott. Truels and strategies for survival. *Scientific Reports*, 9:1–9, 2019.

[5] Adrian P. Flitney and Derek Abbott. Quantum models of parrondo's games. *Physica A*, 324:152–156, 2003.

[6] Junling Hu and Michael P. Wellman. Nash Q-learning for general-sum stochastic games. *Journal of Machine Learning Research*, 4:1039–1069, 2003.

[7] D. Marc Kilgour. The sequential truel. *International Journal of Game Theory*, 4(3): 151–174, 1975.

[8] Clark Kinnaird. Encyclopedia of puzzles and pastimes, 1946.

[9] Michael L. Littman. Markov games as a framework for multi-agent reinforcement learning. In *Machine Learning Proceedings*, pages 157–163, 1994.

[10] Michael L. Littman. Friend-or-foe Q-learning in general-sum games. In *Proceedings of ICML*, 2001.

[11] Richard D. McKelvey and Thomas R. Palfrey. Quantal response equilibria for normal form games. *Games and Economic Behavior*, 10:6–38, 1995.

[12] Lloyd S. Shapley. Stochastic games. *Proceedings of the National Academy of Sciences*, 39 (10):1095–1100, 1953.

[13] Martin Shubik. *Readings in Game Theory and Political Behavior*. Doubleday, 1954.

[14] Martin Shubik. *Game Theory in the Social Sciences*. MIT Press, 1982.

[15] Finbarr Timbers, Edward Lockhart, Marc Lanctot, Martin Schmid, Michael Bowling, and Rémi Munos. Approximate exploitability: Learning a best response. In *NeurIPS Workshop on Offline RL*, 2020.

[16] Raul Toral and Pau Amengual. Distribution of winners in truel games. *AIP Conference Proceedings*, 779:128–131, 2005.

[17] Christopher J. C. H. Watkins and Peter Dayan. Q-learning. *Machine Learning*, 8:279–292, 1992.

[18] Xiaosong Xu. Three-way duel with the weakest contestant: a game theory approach. *International Game Theory Review*, 14(2), 2012.

[19] Martin Zinkevich, Michael Johanson, Michael Bowling, and Carmelo Piccione. Regret minimization in games with incomplete information. In *Advances in Neural Information Processing Systems*, 2007.